LAB EXPERIMENTS – UNIT 2

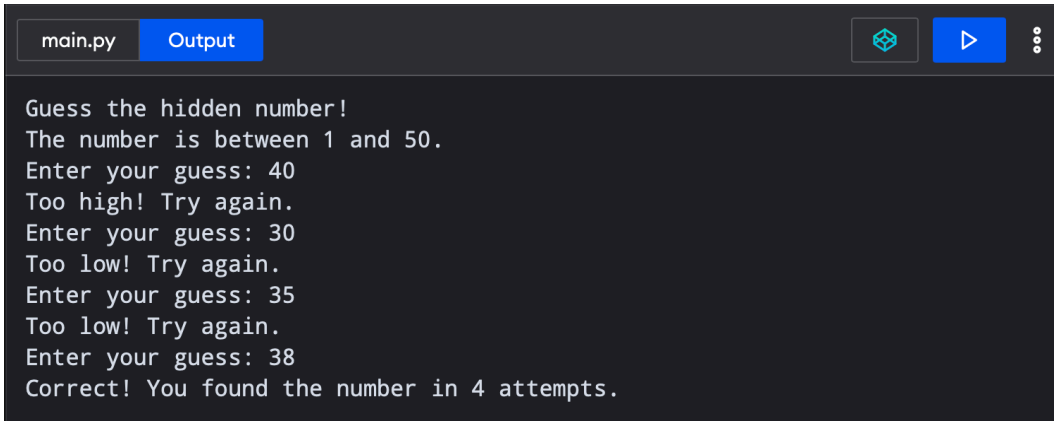
LAB EXERCISE 5: NUMBER GUESSING GAME

CODE:

```
import random

secret = random.randint(1, 50)
attempts = 0
print("Guess the hidden number!")
print("The number is between 1 and 50.")
while True:
    guess = int(input("Enter your guess: "))
    attempts += 1
    if guess < 1 or guess > 50:
        print("Please enter a number between 1 and 50.")
    elif guess == secret:
        print(f"Correct! You found the number in {attempts} attempts.")
        break
    elif guess < secret:
        print("Too low! Try again.")
    else:
        print("Too high! Try again.")
```

OUTPUT:



LAB

EXERCISE 6: PRIME NUMBER AND FACTORIAL PROGRAMS

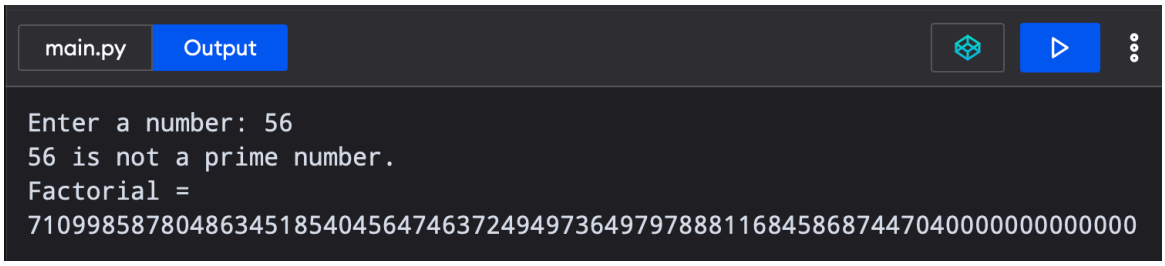
CODE:

```
number = int(input("Enter a number: "))
# Prime number check
if number < 2:
    prime = False
else:
    prime = True
    for value in range(2, number):
        if number % value == 0:
            prime = False
            break

if prime:
    print(number, "is a prime number.")
else:
    print(number, "is not a prime number.")

# Factorial calculation
if number < 0:
    print("Factorial is not available for negative numbers.”)
else:
    factorial = 1
    for value in range(1, number + 1):
        factorial = factorial * value
    print("Factorial =", factorial)
```

OUTPUT:



LAB EXERCISE 7: FIBONACCI SERIES USING RECURSION

CODE:

```
def fib(n):
    if n == 0:
        return 0

    if n == 1:
        return 1
    return fib(n - 1) + fib(n - 2)

terms = int(input("How many Fibonacci numbers do you want? "))

if terms <= 0:
    print("Please enter a positive number.")
else:
    print("Fibonacci sequence:")
    for position in range(terms):
        print(fib(position), end=" ")
```

OUTPUT:

